

Short circuits in vault networks (wash lending)

QuantFi

1 Why use vault shares as collateral

Imagine a vault that lends against its own shares. For instance lending ETH on an ad hoc Morpho market against rcETH or sigmaETH.

A common situation is that where an exogenous LP (not the vault) lends on the back of the vault shares and thereby allows the rcETH investors to leverage their yield with the usual formula $s_V + Mc$.

Since collateral (rcETH) and loan (ETH) share the same unit, the carry $c = s_V - b$ is a spread, not an exchange-rate bet. A loopier at multiplier M (debt over equity) holds $(1 + M)e$ of rcETH against Me of ETH debt, so its net ETH exposure is

$$(1 + M)e - Me = e,$$

its own equity. Leverage amplifies the carry, not the price exposure: the looped part is self-financed in ETH and stays delta-neutral. This is leveraged staking in its native form.

LPs to the market will earn a supply rate ub which has to stay below the original's vault $c = s_V - b > 0$.

(But there is no maturity. So a single large lender is hostage to the loopier; there should be an unwinding agreement, a kind of soft maturity, or callable maturity.)

Assuming the lending market is at steady state b^* , u^* , the joint profitability window reads: (neglecting the variability in exchange rate!)

$$s_V >_{\text{looper}} b^* >_{\text{lender}} s_0/u^*$$

where s_0 is the global ETH lending rate. In particular, the bargaining window is non-empty iff $s_V > s_0/u^*$.

Currently s_V is 3.75%, and $b = 2.12\%$ (both in ETH). Hence $c \sim 1.5\%$.

The loopier's levered yield $A = s_V + Mc$ then reads, at the borrow-rate floor $b = 2.12\%$ ($c = 1.63\%$) and at a more realistic steady state where loopier demand has bid b up toward 3% ($c = 0.75\%$):

M	A ($c = 1.63\%$)	A ($c = 0.75\%$)
5	11.9%	7.5%
10	20.0%	11.3%
15	28.2%	15.0%

all in ETH and genuinely earned, since the loopier keeps ETH exposure only on its equity e .

On the lender side the outside option is the global ETH rate $s_0 = 1.46\%$, so the window opens once $u^* > s_0/s_V = 1.46/3.75 \approx 39\%$. At $u^* = 80\%$ the floor is $s_0/u^* = 1.83\%$, so any

$b^* \in (1.83\%, 3.75\%)$, a band about 190 bp wide, leaves both sides ahead of their alternatives. This is far more robust than a market where the lender’s alternative is a stablecoin rate close to s_V .

The risk migrates with the denomination. Liquidation is governed by the rcETH/ETH exchange rate, not by ETH against the dollar: the collateral impairs only if rcETH underperforms ETH (slashing, a contract loss, a wrapper depeg). The attachment point $\delta_A = 1/(1 + M)$ then measures how far that basis may move before junior equity is wiped; since rcETH/ETH normally carries near-zero volatility, the breach probability $\Phi(-\delta_A/(\sigma\sqrt{\tau}))$ collapses and multipliers up to $\ell/(1 - \ell)$ stay sustainable. The cost is a short-volatility profile: the basis sits flat until a vault-specific event makes it gap, at which point a high- M position exhausts its thin buffer at once and exit liquidity vanishes exactly when it is needed.

Which in turn constrains the max multiplier M debt/equity on the loop side.

1.1 The interest-rebate loop

In other words, shares-as-collateral creates an interest-rebate loop: borrowers pay interest to the vault, then recover part of that interest through their vault shares. This lowers their effective borrow cost, creates self-reinforcing leverage, and can make borrower demand non-monotone in the borrow rate. The note derives the mechanism first in a two-player example, then in a heterogeneous-borrower equilibrium, and closes with a tranche interpretation of the resulting risk transfer.

We are given a vault V where players can deposit a stable S (e.g. USDC) for a share token X (e.g. syrupUSDC). The vault’s strategy is to direct the S s it receives to two venues:

- half goes to another vault L that generates $s = 8\%$ APY (e.g. a leveraged carry position on LSTs)
- half goes into an X/S lending market M where people can deposit- X -and-borrow- S at $b = 3\%$ borrow rate

Now suppose the following scenario with two players P_1 and P_2 . Start with an empty vault.

Step 1

- P_1 deposits 100 S in V for 100 X
- P_2 deposits 100 S in V for 100 X
- V posts 100 S on M
- V posts 100 S on L

Positions:

$$P_1 = 100 X$$

$$P_2 = 100 X$$

$$V = 100 S \text{ (0 lent) on } M@3\% + 100 S \text{ on } L@8\%$$

Yields:

$$\begin{aligned}\text{yield}(V) \text{ on } L &= 100 \times 0.08 = 8 \\ \text{yield}(V) \text{ on } M &= 0 \times 0.03 = 0 \\ \text{yield}(V) &= 8 \\ \text{yield}(P_1) &= \frac{1}{2} \times 8 = 4 \\ \text{yield}(P_2) &= \frac{1}{2} \times 8 = 4\end{aligned}$$

Step 2

- P_2 deposits 100 X as collateral on M
- P_2 borrows 60 S on M (entirely provided by V)
- P_2 deposits 60 S in V for 60 X
- V posts 30 S on M
- V posts 30 S on L

Positions:

$$\begin{aligned}P_1 &= 100 X \\ P_2 &= 160 X - 60 S \\ V &= 130 S \text{ (60 lent) on } M@3\% + 130 S \text{ on } L@8\%\end{aligned}$$

Yields:

$$\begin{aligned}\text{yield}(V) \text{ on } L &= 130 \times 0.08 = 10.4 \quad (\text{external yield}) \\ \text{yield}(V) \text{ on } M &= 60 \times 0.03 = 1.8 \\ \text{yield}(V) &= 12.2 \\ \text{yield}(P_1) &= \frac{100}{260} \times 12.2 = 4.69 \\ \text{yield}(P_2) &= \frac{160}{260} \times 12.2 - 1.8 = 5.71\end{aligned}$$

At that stage we can make a number of remarks:

- If V 's strategy was to put everything on L at step 1, $\text{yield}(V) = 200 \times 0.08 = 16$, so V is leaving money on the table.
- By looping its position, P_2 gets a higher share of the common yield, i.e. $\text{yield}(P_1) < \text{yield}(P_2)$.
- P_2 has changed that somewhat, now V is putting +30 on L .
- P_2 pays 1.8 to borrow but gets back $\frac{16}{26}$ of that, so effective $b_{\text{eff}} = 1.15\% < b$ paid to the rest of the players P_1 .

- $\text{total}X_{\text{in}V}/\text{total}S_{\text{in}V} = 260/260 = 1$.
- $\text{yield}(V) > \text{yield}(P_1) + \text{yield}(P_2) = \text{external yield } 10.4$, which equals the external L yield; the 1.8 from M is an internal transfer from P_2 to all players.
- P_2 has more risk (liquidation), so one can see the premium as a reward for that risk.
- Withdrawals from V are fine as long as M is liquid: right now P_1 can exit because V has 200 instant S at disposal, if P_1 leaves...

Step 3

- P_1 withdraws 100 X
- V pulls 50 S from M
- V pulls 50 S from L

Positions:

$$P_1 = 0 \text{ } X$$

$$P_2 = 160 \text{ } X - 60 \text{ } S$$

$$V = 80 \text{ } S \text{ (60 lent) on } M@3\% + 80 \text{ } S \text{ on } L@8\%$$

Yields:

$$\text{yield}(V) \text{ on } L = 80 \times 0.08 = 6.4$$

$$\text{yield}(V) \text{ on } M = 60 \times 0.03 = 1.8 \quad (\text{assuming constant rate})$$

$$\text{yield}(V) = 8.2$$

$$\text{yield}(P_1) = 0$$

$$\text{yield}(P_2) = \frac{160}{160} \times 8.2 - 1.8 = 6.4$$

Note that P_2 's LTV y_2/x_2 is unchanged.

2 Second attempt

Assume fractions $\varepsilon + \varepsilon' = 1$. Let rates be b (borrow on M) and s (APY on L).

Step 1

- P_1 deposits $x_1 \text{ } S$ in V for x_1 shares X
- P_2 deposits $x_2 \text{ } S$ in V for x_2 shares X
- V posts $\varepsilon(x_1 + x_2) \text{ } S$ on $L@s$
- V posts $\varepsilon'(x_1 + x_2) \text{ } S$ on $M@b$

Yields:

$$\begin{aligned}\text{yield}(V) & \text{ on } L = s\varepsilon(x_1 + x_2) \\ \text{yield}(P_1) & = x_1 s\varepsilon \\ \text{yield}(P_2) & = x_2 s\varepsilon\end{aligned}$$

Step 2

- P_2 posts x_2 X as collateral in M
- P_2 borrows $y_2 < \varepsilon'(x_1 + x_2)$ on M (liquidity constraint; also needs y_2/x_2 lower than maxLTV)
- P_2 deposits y_2 S in V for y_2 X (or has the rate changed?)
- V posts εy_2 S on $L@s$
- V posts $\varepsilon' y_2$ S on $M@b$

Yields: write $T = x_1 + x_2 + y_2$ for the amount of S in V after step 2.

$$\begin{aligned}\text{yield}(V) & = s\varepsilon T + b y_2 \\ \text{yield}(P_1) & = \frac{x_1}{T} (s\varepsilon T + b y_2) \\ \text{yield}(P_2) & = \frac{x_2 + y_2}{T} (s\varepsilon T + b y_2) - b y_2 = (x_2 + y_2) s\varepsilon - b_{\text{eff}} y_2\end{aligned}$$

with:

$$b_{\text{eff}} = \frac{x_1}{T} b$$

the effective rate at which P_2 is borrowing.

Players' net equals external yield:

$$\text{yield}(P_1) + \text{yield}(P_2) = s\varepsilon T$$

Players' incremental gains from looping y_2 (vs. no loop):

$$\begin{aligned}\Delta_1 & = \frac{x_1}{T} (s\varepsilon T + b y_2) - x_1 s\varepsilon = \frac{x_1}{T} b y_2 \geq 0 \\ \Delta_2 & = (x_2 + y_2) s\varepsilon - \frac{x_1}{T} b y_2 - x_2 s\varepsilon = y_2 \left(s\varepsilon - \frac{x_1}{T} b \right)\end{aligned}$$

The loop is attractive to P_2 if $\varepsilon s > \frac{x_1}{T} b$. It may be that $s < b$ (no carry) and yet $\varepsilon s > \frac{x_1}{T} b$ (some notion of effective carry).

Step 3

- P_1 pushes y_1 S from V with $y_1 \in [-x_1, +\infty)$
- V pushes εy_1 to L
- V pushes $\varepsilon' y_1$ to M

Yields: write $T' = x_1 + y_1 + x_2 + y_2$ for the new amount of S in V .

$$\begin{aligned} \text{yield}(V) &= s\varepsilon T' + b y_2 \\ \text{yield}(P_1) &= \frac{x_1 + y_1}{T'} (s\varepsilon T' + b y_2) \\ \text{yield}(P_2) &= \frac{x_2 + y_2}{T'} (s\varepsilon T' + b y_2) - b y_2 = (x_2 + y_2) s\varepsilon - \frac{x_1 + y_1}{T'} b y_2 \end{aligned}$$

If P_1 withdraws everything, $x_1 + y_1 = 0$ and $\text{yield}(P_2) = (x_2 + y_2) s\varepsilon$, so the effective borrowing rate drops to zero (in which case P_2 can redirect the vault allocation to L without V noticing, so to speak— P_2 is a rate maker). If to the contrary P_1 takes a very large position, P_2 is negligible and its rate goes to b (P_2 is rate taker).

3 Heterogeneous borrowers with fixed loanable supply

We now place the wash-lending loop of the previous section inside the borrower equilibrium framework of `morpho.borrower.equilibrium.tex`. There are n borrowers, each characterised by equity e_i , maximum multiplier M_i (risk tolerance), and minimum acceptable APY a_i . Write $E = \sum_{j=1}^n e_j$ for total equity.

The vault V allocates fraction ε to external yield L at rate s and fraction $\varepsilon' = 1 - \varepsilon$ to market M .

- The loanable supply on M is fixed at $\bar{S}$.

The loop

Borrower i deposits e_i of S in V for e_i shares X , posts X as collateral in M , borrows $y_i = m_i e_i$ of S , and re-deposits in V . After all loops:

$$T = E + Y, \quad Y = \sum_j m_j e_j, \quad \varphi = \frac{Y}{T}.$$

Modified APY

Vault income is $s\varepsilon T + rY$. Borrower i 's pro-rata share of vault income minus interest cost gives APY on equity:

$$A_i = (1 + m_i) s\varepsilon + r \frac{Y - m_i E}{E + Y} = \underbrace{(s\varepsilon + r\varphi)}_{c_v} + m_i \underbrace{(s\varepsilon - r(1 - \varphi))}_{\Delta_v}.$$

This has the Morpho form $A = c + m\Delta$, but both the collateral yield $c_v = s\varepsilon + r\varphi$ and the carry $\Delta_v = s\varepsilon - r_{\text{eff}}$ are endogenous: they depend on the borrow rate r and aggregate debt Y through

$$r_{\text{eff}} = r(1 - \varphi) = \frac{rE}{E + Y}.$$

This is the effective borrow cost after the vault rebate (cf. $b_{\text{eff}} = \frac{x_1}{T} b$ from Section 2).

Bang-bang

Since A_i is linear in m_i , the price-taking optimum is still bang-bang: $m_i = M_i$ if $A_i(M_i) \geq a_i$, else $m_i = 0$.

Participation condition (at $Y = \bar{S}$)

Assume the market clears: $Y = \bar{S}$, so $T = E + \bar{S}$ and $\varphi = \bar{S}/(E + \bar{S})$. The condition $A_i(M_i) \geq a_i$ becomes

$$s\varepsilon(1 + M_i) + r \frac{\bar{S} - M_i E}{E + \bar{S}} \geq a_i.$$

Define $\beta_i := (\bar{S} - M_i E)/(E + \bar{S})$ and $\delta_i := a_i - s\varepsilon(1 + M_i)$. The sign of β_i (equivalently, whether $M_i \leq \bar{S}/E$) partitions borrowers into four types:

Type	Condition	Rate sensitivity	Threshold
A	$M_i < \bar{S}/E, \delta_i \leq 0$	always active; higher r helps	—
B	$M_i < \bar{S}/E, \delta_i > 0$	active iff $r \geq \tilde{r}_i := \delta_i/\beta_i$	floor rate
C	$M_i > \bar{S}/E, \delta_i \leq 0$	active iff $r \leq \tilde{\rho}_i := -\delta_i/(-\beta_i)$	ceiling rate
D	$M_i > \bar{S}/E, \delta_i > 0$	never active	—

The boundary $M_i = \bar{S}/E$ is the rate-neutral multiplier: at that leverage, the borrower recaptures exactly the interest it pays through the vault.

Demand curve

Aggregate demand at rate r :

$$D(r) = \underbrace{\sum_{i \in A} e_i M_i}_{D_A} + \underbrace{\sum_{\substack{i \in B \\ r \geq \tilde{r}_i}} e_i M_i}_{D_B(r)} + \underbrace{\sum_{\substack{i \in C \\ r \leq \tilde{\rho}_i}} e_i M_i}_{D_C(r)}.$$

$D_B(r)$ is a step function that increases in r (rate-seekers join); $D_C(r)$ is a step function that decreases in r (standard Morpho exit). Therefore $D(r)$ is generically **non-monotone**.

Equilibrium

The clearing rate r^* satisfies $D(r^*) = \bar{S}$, with the marginal borrower partially filled if D jumps over $\bar{S}$ at a step. Three regimes:

1. $D(0) > \bar{S}$: excess demand at $r = 0$ from Types A+C. The rate rises, Type C borrowers exit. Clearing lies among Type C thresholds $\tilde{\rho}_i$ (standard regime).
2. $D(0) < \bar{S} < D(\infty)$: insufficient demand at low rates. The rate must rise to attract Type B demand. Clearing lies among Type B thresholds $\tilde{r}_i$ (inverted regime).
3. The upward (B) and downward (C) steps interleave: $D(r)$ crosses $\bar{S}$ at multiple values of r . **Multiple equilibria** are possible.

Qualitative differences from the standard model

1. **Self-reinforcing leverage.** As Y rises, φ rises, r_{eff} falls. Each borrower's entry makes others more profitable—a positive externality absent from the standard model.
2. **Inverted rate sensitivity.** Modest-leverage borrowers ($M_i < \bar{S}/E$) are better off when rates are high, because they recapture more interest through the vault than they pay.
3. **Multiple equilibria.** The fixed-point $Y^* = G(Y^*)$, where G maps aggregate debt to consistent demand, can have multiple crossings. A high- Y equilibrium with strong rebate can coexist with a low- Y equilibrium with weak rebate.
4. **Amplified predatory entry.** From `morpho_borrower_equilibrium.tex` §4: a large entrant profits from knocking out incumbents (their exit reduces aggregate debt, lowering the terminal rate). In the wash-lending version this compounds: the entrant's debt also increases φ , boosting the vault rebate. The APY gain from a cascade of k exits is

$$A_0^{(k)} - \hat{A}_0 = m_0 \kappa \sum_{i=1}^k d_i + (1 + m_0) r^* \frac{Y_{\text{post}} - Y_{\text{pre}}}{(E + Y_{\text{post}})(E + Y_{\text{pre}})} E,$$

where the first term is the standard rate-relief channel (linear model, $r(Y) = r_b + \kappa Y$) and the second is the rebate channel from the vault loop.

Application to the Resolv USR incident

USR is the stable S ; the Resolv vault is V ; the compromised mint authority is equivalent to an unbounded shock $\varepsilon' \rightarrow 1$, redirecting all backing to lending with $s\varepsilon \rightarrow 0$. In that limit the carry inverts:

$$\Delta_v = s\varepsilon - r_{\text{eff}} \longrightarrow -r_{\text{eff}} < 0.$$

Every borrower faces negative carry; the bang-bang optimum flips from $m_i = M_i$ to $m_i = 0$. The self-reinforcing leverage that sustained the high- Y equilibrium becomes a self-reinforcing

unwind: each exit reduces φ , raising r_{eff} for survivors, knocking out the next marginal borrower. The cascade terminates only when $Y = 0$ (full deleveraging) or when the rate drops to a level consistent with the remaining active set—but with $s\varepsilon = 0$ no such interior fixed point exists.

This is the blast radius of a mint-authority compromise expressed in the borrower equilibrium language: the entire active set is expelled in a single cascade, and the welfare loss equals $\sum_{i=1}^n e_i [c_v + M_i \Delta_v]$ evaluated at the pre-hack equilibrium.

Implicit tranching

The wash-lending equilibrium admits a structured-finance interpretation that clarifies the risk allocation. Partition participants into two roles:

- **Senior tranche (lenders).** Players who deposit S in V but do not loop. Their yield is $s\varepsilon$ —the vault’s external rate, independent of the borrow market. They bear no liquidation risk and are first in line for the vault’s assets.
- **Junior tranche (loopers).** Players who loop S back through M . Their yield is $c_v + M_i \Delta_v$, which exceeds the senior yield when $\Delta_v > 0$ (positive carry) but falls below it—and can go negative—when rates spike or the backing collapses. They are levered, bear liquidation risk, and absorb first losses.

The borrow rate r^* is the endogenous price of risk transfer between the two tranches: it determines both the senior coupon $s\varepsilon + r^* \varphi$ (boosted by interest income) and the junior spread $\Delta_v = s\varepsilon - r^*(1 - \varphi)$.

Bifurcation of the tranche split. The multiple equilibria from the clearing analysis above acquire a direct structural interpretation: at the same pool configuration (same E , same $\bar{S}$, same borrower population), the market can settle into

1. a **thin junior** equilibrium: low Y , weak rebate, high effective borrow cost r_{eff} —only the most risk-tolerant borrowers lever up, and most capital sits in the senior tranche earning $\approx s\varepsilon$;
2. a **thick junior** equilibrium: high Y , strong rebate, low r_{eff} —the self-reinforcing leverage channel pulls in modest-leverage borrowers (Type B), the junior tranche dominates the capital structure, and the senior tranche earns a higher coupon from the large interest flow.

The transition between these states is discontinuous: a small shock to $\bar{S}$ or s can flip the pool from one regime to the other, producing a sudden reallocation of risk that looks like a run on the junior tranche (borrowers exit en masse) or a leverage spiral (new borrowers pile in).

Comparison with traditional CDOs. In a standard CDO the tranche boundaries (attachment/detachment points) are fixed at issuance. Here they are endogenous: the size of the junior tranche (Y) and its compensation (Δ_v) are equilibrium objects determined by

the borrower population and the vault's allocation rule ε/ε' . Governance changes to ε (e.g. Sky adjusting the DSR or the Spark allocation) are analogous to restructuring the waterfall mid-life—something impossible in traditional structured finance but routine in DeFi.

Application. Spark/Sky is the canonical example: DAI lenders are senior (yield = DSR $\approx s$), DAI borrowers who loop through Spark are junior (leveraged carry on collateral yield minus effective borrow cost). The DSR is the governance lever that reprises the tranche split—raising it simultaneously increases senior yield and compresses junior carry, potentially triggering the thin-junior equilibrium.

Tranche attachment points

The analogy with structured finance becomes quantitative once we compute the collateral-loss threshold at which junior is wiped out and losses begin hitting senior.

Setup. At equilibrium the junior tranche (loopers) holds aggregate collateral value

$$X_J = \sum_{i \in \text{active}} (1 + M_i) e_i = E_J + Y,$$

where $E_J = \sum_{i \in \text{active}} e_i$ is junior equity and $Y = \bar{S}$ is total debt. The senior tranche holds $E_S = E - E_J$ (non-looping deposits). Total vault assets are $T = E + Y$ in S -terms, backed by collateral worth X_J at current prices plus the vault's external position εT on L .

Collateral shock. Suppose the collateral price p_X drops by a fraction δ , so collateral is worth $(1 - \delta) X_J$. Borrower i is liquidated when their LTV exceeds the liquidation threshold ℓ :

$$\frac{m_i e_i}{(1 - \delta)(1 + m_i) e_i} > \ell \quad \Longleftrightarrow \quad \delta > 1 - \frac{m_i}{\ell (1 + m_i)}.$$

Define the individual liquidation threshold

$$\delta_i^* := 1 - \frac{m_i}{\ell (1 + m_i)} = 1 - \frac{M_i}{\ell (1 + M_i)}$$

(using $m_i = M_i$ at the bang-bang equilibrium). Borrowers with higher leverage are liquidated first: δ_i^* is decreasing in M_i .

Junior wipeout (attachment point). As δ increases past each δ_i^* , the corresponding borrower is liquidated. If liquidation recovers $(1 - \delta)(1 + M_i) e_i$ against debt $M_i e_i$, the shortfall per borrower is

$$L_i(\delta) = \max(0, M_i e_i - (1 - \delta)(1 + M_i) e_i) = e_i \max(0, \delta(1 + M_i) - 1).$$

The junior tranche absorbs losses up to its total equity E_J . The **attachment point** δ_A is the smallest shock at which cumulative losses exhaust junior equity:

$$\sum_{i \in \text{active}} L_i(\delta_A) = E_J.$$

For $\delta > \delta_A$, losses spill into the senior tranche (vault depositors who did not loop).

Closed form (homogeneous case). If all active borrowers have the same multiplier M , then $E_J = Y/M$, $X_J = (1 + M) E_J$, and the attachment point simplifies to

$$\delta_A = \frac{1}{1 + M} (1 + M(1 - \ell)) = 1 - \frac{M\ell}{1 + M}.$$

At Spark-like parameters ($\ell = 0.83$, $M = 4$): $\delta_A = 1 - \frac{4 \times 0.83}{5} = 1 - 0.664 = 0.336$, meaning a 33.6% collateral drop wipes out junior. At Morpho’s high-LLTV regime ($\ell = 0.945$, $M = 15$): $\delta_A = 1 - \frac{15 \times 0.945}{16} = 1 - 0.886 = 0.114$, only 11.4%.

Detachment point. The **detachment point** δ_D is the shock at which total vault losses equal $E_J + E_S = E$ (both tranches wiped, only debt remains). In the homogeneous case:

$$\delta_D = 1.$$

Since collateral can’t go below zero, $\delta_D = 1$ always—but the senior tranche’s loss grows linearly for $\delta \in [\delta_A, 1]$.

Endogenous attachment. Unlike a fixed CDO, δ_A depends on the equilibrium leverage M^* (or the distribution of M_i ’s in the heterogeneous case), which itself depends on $\bar{S}$, s , ε , and the borrower population. A governance change that shifts the equilibrium from thin-junior to thick-junior simultaneously lowers the attachment point—more leverage means the loss-absorption buffer shrinks. This is the quantitative content of the “endogenous CDO” analogy: the waterfall reshapes itself in response to market conditions and governance actions.

Volatility-adjusted loss probability. If collateral returns are normally distributed with annualized volatility σ and the position horizon is τ (e.g. the time to liquidation oracle update), the probability of hitting the attachment point is

$$\Pr[\delta > \delta_A] = \Phi\left(\frac{-\delta_A}{\sigma\sqrt{\tau}}\right),$$

where Φ is the standard normal CDF.

For ETH ($\sigma \approx 0.80$) over a 1-day horizon ($\tau = 1/365$), the probability of a 33.6% drop is negligible; over a 7-day horizon it is still small but non-trivial for tail events.

Choice of horizon τ . The horizon τ should reflect the lender’s worst-case time to exit under the stress scenario being priced: $\tau = \tau_{\text{detect}} + \tau_{\text{exit}}$, where τ_{detect} is the delay before the lender recognises the drawdown, and τ_{exit} is the time needed to withdraw supply from the pool. On Morpho Blue, τ_{exit} depends on available liquidity: if utilisation is high, only a small fraction of supply can be redeemed instantly, and a depeg event triggers correlated withdrawals that further reduce liquidity—a bank-run dynamic that makes τ_{exit} endogenous to the crisis. Crucially, the sensitivity of the fair spread to τ is highly nonlinear and market-dependent. For thick-buffer markets ($\delta_A \gtrsim 50\%$, e.g. ETH/BTC-collateral pools), increasing τ from 7 to 30 days barely moves the breach probability, because $\delta_A/(\sigma\sqrt{\tau})$ remains deep in the tail. For thin-buffer stablecoin loops ($\delta_A \approx 5\text{--}15\%$), the same change can push the breach probability from negligible to order-one, causing the fair spread to explode. A single global τ therefore masks systematic underpricing of the most fragile markets; the disciplined

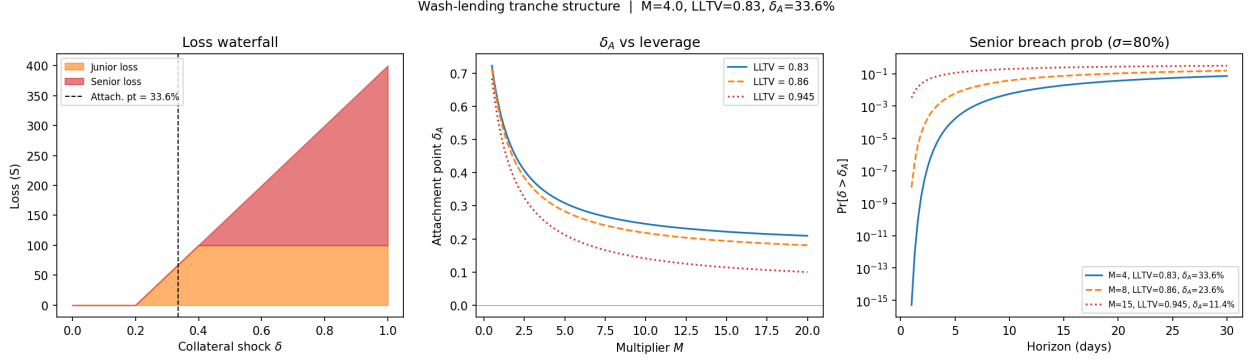


Figure 1: Loss waterfall and attachment-point sensitivity from `wash.tranche.py`. Left: junior absorbs losses until $\delta = \delta_A$, senior breaches thereafter. Center: attachment point versus leverage across Spark/Maker-style LLTV regimes. Right: senior-loss probability under ETH-like volatility ($\sigma = 80\%$).

approach is to assign a per-market or per-asset-class horizon—e.g. $\tau = 7$ d for liquid crypto-collateral pools and $\tau = 30$ d for stablecoin loops where exit liquidity evaporates precisely when it is most needed.

The companion script `wash.tranche.py` plots the full waterfall and loss distributions for various parameter regimes.

4 Case study: the Ethena (USDe) self-lending estimate

On 20 May 2026 the account `@yieldsandmore` (YAM) published a thread¹ quote-tweeting Ethena’s updated transparency page. Its claim: of the $\approx \$4.4$ B backing the synthetic dollar USDe, an estimated $\$1.33$ B—about 30%—is “lending against itself.” The estimate rests on a pro-rata rule applied market by market:

$$\text{self-lending} = \sum_k \varphi_k B_k,$$

where B_k is gross borrowing against USDe/sUSDe collateral in market M_k and φ_k is Ethena’s share of supplied liquidity there. This section maps that thread onto the wash-lending model and then asks what a static estimate of this kind cannot see.

The data

The thread’s market-by-market breakdown (all figures in USD millions):

¹<https://x.com/yieldsandmore/status/2057150511883710628>

Venue	Market	Ethena supplied	USDe+sUSDe borrowed	Lender share	Est. self- lending
Aave	Plasma USDT0	368.0	556.0	47.4%	263.3
Aave	Mantle USDT0	168.0	58.1	100.0%	58.1
Aave	Ethereum USDtb	140.0	46.7	82.5%	38.5
Aave	MegaETH USDm	550.0	398.9	91.2%	363.8
Morpho	Base Prime USDC	252.9	0.0	54.3%	0.0
Morpho	Ethereum USDtb	105.0	105.3	99.7%	105.0
Kamino	Sentora USDG	250.0	250.0	100.0%	250.0
Juplend	Bitwise USDG	251.0	251.0	100.0%	251.0
Total		2084.9	1666.0		1329.7

The gross figure—total borrowing collateralised by USDe/sUSDe—is $\approx \$1.67\text{B}$; the pro-rata attribution discounts it to $\approx \$1.33\text{B}$, a discount of $\$336\text{M}$ representing leverage funded by lenders other than Ethena.

Mapping onto the model

The correspondence with the vault network of the preceding sections is:

- V — Ethena: the mint contract together with the reserve manager that allocates the backing.
- S — the loanable stablecoins actually borrowed in the table (USDC, USDT0, USDtb, USDG).
- X — USDe and its derivatives sUSDe and PT-sUSDe: the share-side tokens posted as collateral.
- L — the delta-neutral basis trade (spot crypto hedged with short perpetuals) that produces the external yield s .
- M_k — the family of USDe/sUSDe-collateral lending markets, one per row of the table.

The key identification is methodological. YAM’s per-market multiplier “Ethena’s share of supplied liquidity” is precisely a supply-share attribution: it apportions the debt of market M_k to V in proportion to V ’s lending into M_k . Summed over markets, YAM’s $\$1.33\text{B}$ is therefore an empirical estimate of the V -funded part of aggregate looped debt Y . The model’s $\varphi = Y/T$ is exactly the quantity the thread is reaching for: the fraction of the vault that is financing leverage on its own shares.

One accounting point follows immediately. The headline $\$4.4\text{B}$ is “backing,” i.e. total assets held by V ; in the model that is $T = E + Y$, including the looped redeposits. Backing and leverage are not disjoint: part of the $\$4.4\text{B}$ is USDe minted with stablecoins that were themselves borrowed against USDe. This is a TVL double-count—here measured rather than asserted.

Quantification

Take $T = \$4.4\text{B}$ and $Y \in [\$1.33, \$1.67]\text{B}$ (pro-rata vs. gross). Then

$$\varphi = \frac{Y}{T} \in [0.30, 0.38].$$

The remaining model quantities require rates and a leverage level, which the thread does not give. We use the following illustrative values (not Ethena data): basis yield $s = 9\%$; vault allocation to L of $\varepsilon = 0.53$ (the complement of the $\$2.08\text{B}/\$4.4\text{B} \approx 0.47$ supplied into markets); borrow rate $r = 6\%$; representative looper multiplier $M = 6$; PT-sUSDe liquidation threshold $\ell = 0.915$. At $\varphi = 0.30$:

$$\begin{aligned} r_{\text{eff}} &= r(1 - \varphi) = 0.70 \times 6\% = 4.2\% && \text{(loop rebates 30\% of interest)} \\ c_v &= s\varepsilon + r\varphi = 4.77\% + 1.80\% = 6.57\% \\ \Delta_v &= s\varepsilon - r(1 - \varphi) = 4.77\% - 4.20\% = +0.57\% && \text{(thin positive carry)} \end{aligned}$$

A representative looper earns $A = c_v + M\Delta_v \approx 6.6\% + 6 \times 0.57\% \approx 10.0\%$, against $c_v \approx 6.6\%$ for a non-looping holder—the loop roughly adds half again the base yield, on a carry of only 57 basis points.

The leverage cap is $\ell/(1 - \ell) \approx 10.8$, so $M = 6$ is well within range. In the recursive-loop convention, the two collateral-shock thresholds—the liquidation onset δ_{liq} and the looper-equity wipeout δ_A —are

$$\delta_{\text{liq}} = 1 - \frac{M}{\ell(1 + M)} = 1 - \frac{6}{0.915 \times 7} \approx 6.3\%, \quad \delta_A = \frac{1}{1 + M} = \frac{1}{7} \approx 14.3\%.$$

A USDe price decline of $\approx 6\%$ begins liquidating the representative looper; a decline of $\approx 14\%$ exhausts looper equity and pushes losses into the senior tranche—the non-looping USDe holders, i.e. the backing itself.

What the static estimate misses

The $\$1.33\text{B}$ is a stock, measured at one instant. Three features of Ethena, all invisible to a pro-rata snapshot, are visible to the model.

(i) The collateral is a peg. In the model the collateral shock δ is exogenous and “structural.” Here $X = \text{USDe}$, whose price is pinned at $\$1$ by design; a collateral shock is a depeg. The attachment arithmetic above shows the buffer is thin—liquidations begin near a 6% deviation. YAM’s estimate quantifies how much circular leverage exists; it says nothing about the flow that a depeg sets off once $\delta > \delta_{\text{liq}}$.

(ii) The external yield can invert. L is a perpetual-funding basis trade, not a staking carry; s is a funding rate. When funding inverts, s falls and may go negative, so $\Delta_v = s\varepsilon - r_{\text{eff}}$ turns negative and the bang-bang optimum flips every looper from $m_i = M_i$ to $m_i = 0$. This is the Resolv USR limit ($s\varepsilon \rightarrow 0$) discussed earlier—reached not by a mint-authority compromise but by an ordinary change in funding regime. A snapshot taken while $\Delta_v > 0$ reads the loop as benign precisely because it is, for now.

(iii) φ is not a scalar. Each market M_k has its own lender share φ_k and its own LLTV ℓ_k , hence its own attachment point δ_A^k . The four rows where Ethena supplies essentially all the liquidity—Mantle USDT0, Morpho Ethereum USDtb (99.7%), Kamino Sentora USDG, Juplend Bitwise USDG—are fully circular: V is the only lender, the interest rebate within that market is total, and there is no third-party capital to absorb a shock. The aggregate \$1.33B hides the fact that these markets carry the highest effective φ_k and unwind first; the cascade is ordered by the distribution of δ_A^k , not by a single number.

In short, the pro-rata estimate answers a stock question—how much of the backing is V financing its own leverage. The model reframes it as a set of flow questions: how thin is the carry Δ_v , how endogenous and self-reinforcing is φ , and how fast does the stock unwind once a depeg crosses δ_{liq} . Those are the natural sequel to a static self-lending estimate.

5 About Paul’s tweet

I would separate the analysis in two separate sections.

Suppose the LLTV of a market B/A (collateral B , loanable A) is 0.945. One can take a position with collateral deposit $(M + 1)x$ and borrow amount Mx , both counted in USD value (according to the oracle), as long as

$$M \leq \frac{\text{LLTV}}{1 - \text{LLTV}} \approx 17.2.$$

Suppose we take $M = 15$. Then the ratio borrowAmount/positionValue = $Mx/x = 15$; we can call this leverage (even if some reserve the term for $M + 1$); leverage zero corresponds to just depositing collateral, no borrow.

Now we can define TVL as AmountCollateralSupplied, and not subtract the amount borrowed. With that convention, taking the position increases the TVL by $(M + 1)x = 16x$. If you have 1M you can inflate the TVL by 16M.

We can also define TVL as AmountCollateralSupplied + AmountLoanableLent = $(2M + 1)x$, in our case = $31x$.

Now if we aggregate the lender (issuer role) and the looper (user role):

1. User deposits x of B , mints x of A .
2. Issuer deposits x of B in A/B market.
3. User builds position $(M + 1)x$ of A , Mx of B (debt).

6 Revision notes

These are editorial and mathematical notes to incorporate in a later revision. They are intentionally appended rather than folded into the main text, so the current argument remains unchanged.

1. Collateral convention

The current draft mixes two different looping conventions. In the initial numerical example, P_2 starts with $100X$, posts that $100X$ as collateral, borrows $60S$, and deposits the borrowed $60S$ back into V for another $60X$. The balance sheet is then written as

$$P_2 = 160X - 60S.$$

That is economically correct as a net position, but it is ambiguous as a lending-market position. Did P_2 pledge only the first $100X$, or did P_2 also pledge the newly minted $60X$?

If only the first $100X$ is collateral, the lending-market LTV is

$$\frac{60}{100} = 60\%.$$

If the full $160X$ is collateral, the lending-market LTV is

$$\frac{60}{160} = 37.5\%.$$

These are different models.

The later tranche and TVL sections use the recursive-loop convention: initial equity is e_i , debt is $m_i e_i$, and total vault shares/collateral after looping are $(1 + m_i)e_i$. Under that convention the lending-market LTV is

$$\frac{m_i e_i}{(1 + m_i)e_i} = \frac{m_i}{1 + m_i},$$

so the LLTV constraint is

$$\frac{m_i}{1 + m_i} \leq \ell \quad \Longleftrightarrow \quad m_i \leq \frac{\ell}{1 - \ell}.$$

The paper should choose one convention and carry it throughout. Since the final section on Paul's tweet uses the recursive convention, the cleanest revision is probably to rewrite the earlier examples in that language:

$$\begin{aligned} e_i &= \text{initial equity,} \\ m_i e_i &= \text{borrowed amount,} \\ (1 + m_i)e_i &= \text{total collateral/shares after looping.} \end{aligned}$$

This avoids using M sometimes as borrow-against-original-equity and sometimes as recursive debt/equity multiplier.

2. Senior yield statement

The current tranche interpretation says that non-looping depositors earn $s\varepsilon$ and that their yield is independent of the borrow market. That is not consistent with the earlier model. The heterogeneous borrower section derives

$$A_i = c_v + m_i \Delta_v, \quad c_v = s\varepsilon + r\varphi, \quad \Delta_v = s\varepsilon - r(1 - \varphi).$$

A non-looper has $m_i = 0$, hence

$$A_i = c_v = s\varepsilon + r\varphi.$$

So senior/non-looping depositors do receive borrow-market income through the vault. They do not only receive the external L yield. The actual distinction is:

$$\text{senior/non-looper yield} = c_v,$$

while

$$\text{junior/looper yield} = c_v + m_i\Delta_v.$$

Both groups receive the base vault return c_v . Loopers additionally pay borrowing costs and lever the base return, so their incremental economics are governed by

$$\Delta_v = s\varepsilon - r(1 - \varphi).$$

That is the more precise tranche split: seniors get the base vault return; juniors lever it and are exposed to whether incremental carry is positive or negative.

3. Attachment-point math

The attachment-point section needs a careful rewrite. It currently defines

$$L_i(\delta) = \max(0, M_i e_i - (1 - \delta)(1 + M_i)e_i).$$

This is not the loss absorbed by the junior tranche. It is the lender/senior shortfall after collateral value has already fallen below debt. In other words, it starts counting losses after junior equity is gone.

For a homogeneous recursive loop, let

$$\text{collateral} = (1 + M)e, \quad \text{debt} = Me.$$

After a collateral price drop δ , collateral is worth

$$(1 - \delta)(1 + M)e.$$

The looper's remaining equity is

$$(1 - \delta)(1 + M)e - Me = [1 - \delta(1 + M)]e.$$

Junior equity is wiped when this equals zero:

$$1 - \delta(1 + M) = 0 \quad \Longleftrightarrow \quad \delta_A = \frac{1}{1 + M}.$$

Thus if $M = 4$, junior is wiped at $1/5 = 20\%$. If $M = 15$, junior is wiped at $1/16 = 6.25\%$. These are the clean attachment-point numbers when attachment means “senior losses begin” absent liquidation.

The liquidation threshold is separate. If the liquidation LLTV is ℓ , liquidation occurs when

$$\frac{Me}{(1 - \delta)(1 + M)e} > \ell.$$

Solving gives

$$\delta_{\text{liq}} = 1 - \frac{M}{\ell(1 + M)}.$$

For $\ell = 0.83$ and $M = 4$,

$$\delta_{\text{liq}} = 1 - \frac{4}{0.83 \cdot 5} \approx 3.6\%.$$

Liquidation therefore triggers long before junior equity is fully wiped, which is the intended function of liquidation.

The current formula

$$1 - \frac{M\ell}{1 + M}$$

is neither the liquidation trigger nor the equity wipeout point under the recursive definitions above. A cleaner revision should separate:

$$\delta_{\text{liq}} = 1 - \frac{M}{\ell(1 + M)} \quad (\text{liquidation starts}),$$

$$\delta_A = \frac{1}{1 + M} \quad (\text{junior equity is gone and senior loss begins absent liquidation}),$$

and

$$L_{\text{senior}}(\delta) = \max(0, Me - (1 - \delta)(1 + M)e) \quad (\text{senior shortfall after attachment}).$$

4. Fixed loanable supply versus vault allocation

The heterogeneous section currently combines two assumptions that are not automatically compatible. First, the vault allocates a fixed fraction ε to L and $\varepsilon' = 1 - \varepsilon$ to M . Second, the loanable supply on M is taken to be fixed at $\bar{S}$, and the clearing condition is written as

$$Y = \bar{S}.$$

If M supply comes from the vault's allocation rule, then the supply is not exogenous. It is

$$S_M(Y) = \varepsilon'(E + Y).$$

If all supplied funds are borrowed at clearing, then

$$Y = \varepsilon'(E + Y),$$

so

$$Y(1 - \varepsilon') = \varepsilon'E \quad \Longleftrightarrow \quad Y = \frac{\varepsilon'}{\varepsilon}E.$$

The model should therefore choose one of two clean formulations.

Fixed external supply. The lending market has exogenous loanable supply $\bar{S}$. The vault may own some or all of that supply, but $\bar{S}$ is treated as fixed for the equilibrium exercise. In that case the fixed-fraction allocation language should be softened in this section.

Endogenous vault supply. Loanable supply is

$$S_M(Y) = \varepsilon'(E + Y),$$

and borrowing must satisfy

$$Y \leq S_M(Y).$$

Then clearing is a fixed point involving both borrower demand and the vault allocation rule. This is more faithful to the wash-lending loop, but it is also more involved.

5. Price-taking versus strategic loopers

The bang-bang result is correct as a price-taking benchmark, but the assumption should be stated explicitly. Since

$$A_i = c_v + m_i \Delta_v$$

is linear in m_i , the optimum is $m_i = M_i$ or $m_i = 0$ if borrower i treats

$$r, \quad \varphi, \quad c_v, \quad \Delta_v$$

as fixed.

But a large looper changes aggregate debt Y , hence changes

$$\varphi = \frac{Y}{E + Y}, \quad r_{\text{eff}} = r(1 - \varphi), \quad \Delta_v = s\varepsilon - r(1 - \varphi).$$

So a large looper is not simply choosing leverage at a given spread; the looper changes the spread by looping. This is already visible in the two-player example: when P_1 exits, P_2 becomes the residual owner of the vault and recaptures all interest paid to the vault, so the effective borrow cost falls to zero.

The revision should split the analysis into:

atomistic benchmark: r and φ fixed,

where the bang-bang result applies, and

strategic borrower case: m_i affects Y, φ , and possibly r ,

where the objective is nonlinear and the bang-bang conclusion does not automatically follow.

6. Suggested structure

The draft currently reads like research notes, jumping from the toy example to the general model, the Morpho equilibrium, Resolv, the CDO analogy, liquidation math, and Paul's tweet. For circulation, the main line should probably be narrower.

A clearer structure would be:

1. Start with the toy example. The goal is to make the reader see the mechanism numerically: P_2 borrows from the vault, pays interest to the vault, and recaptures part of that interest through vault ownership.
2. Generalize to the two-player model. Derive the central identity

$$b_{\text{eff}} = b \frac{x_1}{T}.$$

3. Move to many borrowers. Define E , Y , and φ , then derive

$$A_i = c_v + m_i \Delta_v.$$

4. Discuss equilibrium consequences: non-monotone demand, inverted rate sensitivity, multiple equilibria, and the positive externality from aggregate leverage.
5. Introduce the tranche interpretation as an economic analogy, not as literal legal seniority.
6. Put liquidation and loss-threshold math after the tranche interpretation, separating liquidation trigger, junior wipeout, senior shortfall, and exit-liquidity horizon.
7. Move the note on Paul’s tweet and TVL inflation conventions to an appendix or separate closing note.

7. Draft markers and portability

A few local draft markers should be resolved before sharing the PDF.

The parenthetical “or has the rate changed?” in Step 2 should be replaced by the actual modeling assumption. If shares are minted at current NAV and there has been no accrued yield or loss yet, then $y_2 S$ mints $y_2 X$; otherwise it mints

$$\frac{y_2}{\text{NAV}}$$

shares.

The phrase “so to speak” in the Step 3 discussion should be replaced by a precise statement: when P_1 exits, P_2 becomes the residual owner of the vault and recaptures all interest paid to the vault.

The former hard-coded style path `\usepackage{/Users/V/icd/sty/vmacros}` made the document non-portable. The only needed macro from that file was `\ttt`, now defined locally in the preamble. If more convenience macros are introduced later, either define them locally or vendor `vmacros.sty` into the repo.

Finally, the abstract added at the front can be further refined around this core statement:

A vault that lends against its own shares creates an interest-rebate loop: borrowers pay interest to the vault, then recover part of that interest through their vault shares. This lowers their effective borrow cost, creates self-reinforcing leverage, and can make borrower demand non-monotone in the borrow rate.